Medical X-Ray Segmentation: A Combined Theoretical Approach

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Abstract:

Background: Image segmentation is an incredibly useful computer vision technique for understanding a digital image's composition. Many researchers utilize a traditional segmentation pipeline: over-segmentation, region adjacency graph (RAG), and agglomerative clustering on a similarity function over neighbors. Likewise, Delaunay Triangulation is a popular geometric technique, comprising various properties, that has been utilized in feature extraction and the segmentation of digital images. I combine these popular techniques into a combined image segmentation framework.

Methods: My algorithm works in three steps: 1) Delaunay Triangulation over the digital image, 2) an iterative variance-based refinement of the triangulation (crafting atomic, labeled segments), and 3) agglomerative clustering over the atomic segments (the RAG), performing a region-growing step.

Results: I prove the algorithm maintains the properties of the initial triangulation throughout each step, demonstrating its termination in finite steps and backbone. More proofs and empirical testing are required to further prove the algorithm, and its potential downstream effects in classification.

Introduction/Background:

Previous Research:

Part 1 of my algorithm relies on a Delaunay Triangulation as a sort of 'prior' segmentation to be iteratively refined later in the algorithm. This decision was not made ignorantly. Previous research in image segmentation, and feature extraction from digital images, has been conducted with Delaunay triangles. For example, a study from 2016 utilized Delaunay Triangulation in detecting important regions for melanoma.[1] Another study created a novel algorithm for superpixel segmentation, which demonstrated the potential for Delaunay-based segmentation methods.[2]

Delaunay Triangulation:

As mentioned above, the initial over-segmentation is truly the backbone of my algorithm. As such, it is prudent to delve into some statistical properties of Delaunay triangles, which make proving the algorithm an easier endeavor.

Motivation - The Voronoi Diagram:

The Voronoi Diagram is the planar subdivision of n points into convex polygons around sets of "generator" points (or sites). Every point within these regions (Voronoi cells) are closer to these sites than any other Voronoi cell. The dual of the Voronoi

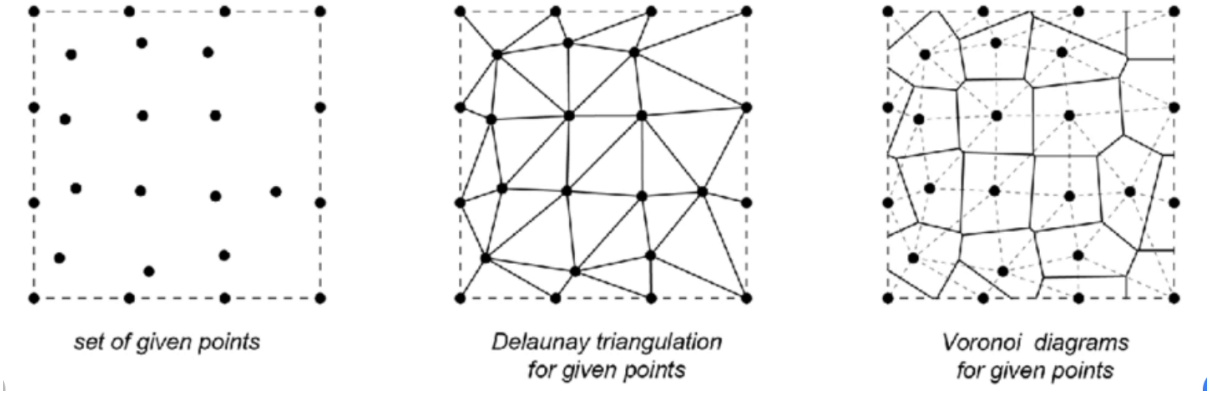


Figure 1: Voronoi Diagram and Delaunay Triangulation Demonstration [3]

Diagram is called a Delaunay Triangulation, where for each site in the diagram, we create a vertex and add edges between vertices for neighboring regions. The relation to these two concepts can be shown in Figure 1.

Properties:

The resulting Delaunay Triangulation contains various properties[4] (See reference 4 for proofs of all these properties):

Convex Hull Property: Edges forming the convex hull of the original points set are also edges of the Delaunay Triangulation. i.e. a union of all triangles in the Delaunay Triangulation is the convex hull of the original set of points.

Circumcircle Property: The circumcircle of any triangle in the Delaunay Triangulation is "empty" - The interior contains no sites of P.

Empty Circle Property: Two sites, pi, pj are connected by an edge in the triangulation iff there is an empty circle passing through the two.

Closest Pair Property: The closest pair of sites in the plane are neighbors in the Delaunay Triangulation.

Euclidean Minimum Spanning Tree: Computing first the Delaunay Triangulation of a point set P and then computing the minimum spanning tree (MST) of the triangulation. The MST of a point set P in any dimension is a subgraph of the Delaunay Triangulation.

The Algorithm:

Algorithm Pt.1: Delaunay Triangulation

Input: A Normalized Grayscale Image I: $\mathbf{\Omega} \rightarrow [\mathbf{0}, \mathbf{1}]$, where $\mathbf{\Omega} \subset \mathbb{Z}^{\mathbf{2}}$ is the finite set of pixel coordinates and I(x,y) denotes the pixel intensity at pixel (x,y)

Output: Finite set of triangles $\mathbf{T} = \{\mathbf{T_1}, \mathbf{T_2}, ..., \mathbf{T_N}\}$ which forms a Delaunay Triangulation of the point set $\mathbf{P} = \mathbf{\Omega}$, and, for each triangle $\mathbf{t} \in \mathbf{T}$, a set of pixels $\mathbf{p_t}$

a. Compute the Delaunay Triangulation of P, obtaining a finite set of triangles $\mathbf{T}$

b. For each triangle $\mathbf{t} \in \mathbf{T}$, the pixels within $\mathbf{t}$, $\mathbf{p_t}$, are defined as:

$$\mathbf{p_t} = \{(\mathbf{x}, \mathbf{y}) \in \mathbf{\Omega};\ (\mathbf{x}, \mathbf{y}) \in \mathbf{t}\}$$

c. Return $\mathbf{T}$ and $\{\mathbf{p_t}\}_{\mathbf{t} \in \mathbf{T}}$

Lemma 1. ***The Delaunay Triangulation forms a Partition of the space P***

Proof. Based on the properties of Delaunay Triangles mentioned in the "Introduction/Background" section, and by the definition of a partition (mutually exclusive and exhaustive), the triangulation clearly satisfies this requirement. The triangulation is crafted such that the circumcircle of every triangle contains no other sites from P in its interior. i.e no two triangles can overlap internally. This is the empty circle and circumcircle properties mentioned above.

Similarly, the triangulation is derived such that the entire region is subdivided into triangles that satisfy the empty circle properties. So, as we can see, the union of all Delaunay triangles forms the convex hull of the original point set.

Since we have shown how the initial triangulation creates non-overlapping triangles that together form the convex hull of P, the triangulation thus forms a partition of the space. □

Algorithm Pt.2: Triangle Refinement

Input: Initial triangulation $\mathbf{T}$, pixel-sets $\{\mathbf{p_t}\}_{\mathbf{t} \in \mathbf{T}}$, a variance threshold $\mathbf{v_{threshold}}$, and a minimum pixel count $\mathbf{n_{min}} = \mathbf{2}$

Output: A refined triangulation (**T'**) over image $\mathbf{I}$, and labels for each triangle $\mathbf{l_t}$; $\forall \mathbf{t} \in \mathbf{T'}$

Variance Condition: Let each pixel-set $\{\mathbf{p_t}\}_{\mathbf{t} \in \mathbf{T}}$ from the initial triangulation contain pixel information (true by definition of $\mathbf{p_t}$), then each pixel-set contains a variance of intensities. Denoting the variance of the pixel-set as $\mathbf{v_{p_t}}$, a triangle is considered homogeneous (passing the variance condition) iff $(\mathbf{v_{p_t}} \leq \mathbf{v_{threshold}}) \vee (|\mathbf{p_t}| \leq \mathbf{n_{min}})$

Refinement: Let **T'** be the set of triangles that pass the *Variance Condition*

a. For $\mathbf{t} \in \{\mathbf{1}, ..., \mathbf{N}\}$ in $\mathbf{T}$:

b. If the Variance Condition is met, append t to T'

c. Else:

- Recursively split until all sub-triangles are homogeneous and append to T'
- i.e. split t into two sub-triangles by connecting the centroid of t to the midpoint of the longest edge
- Calculate $\mathbf{v_{p_t}}$ for both halves of the old triangle (t) and continue splitting by the centroid until the condition is met
- Append each sub-triangle to T' once the *Variance Condition* is met

d. For t in T':

- $n_t = |p_t|$
- $l_t = \frac{1}{n_t} \sum_{(x,y) \in p_t} I(x, y) \in [0, 1]$

e. Return **T'** and the labels $\{\mathbf{l_t}\}_{t \in T'}$

The *Variance Condition* will enact throughout the algorithm until a non-homogeneous triangle ($\mathbf{v_{p_t}} < \mathbf{v_{threshold}}$) is divided entirely into sub-triangles whose pixel-set variance is less than $v_{threshold}$ or the number of pixels within each triangle $p_t \leq 2$

Lemma 2. ***Refinement Termination***
The refinement terminates in a finite number of steps

Proof. With a fixed minimum pixel count $\mathbf{n_{min}} = \mathbf{2}$, any triangle, t, can be refined iff it contains more than $\mathbf{n_{min}}$ pixels and fails homogeneity, $\mathbf{v_{p_t}} > \mathbf{v_{threshold}}$. i.e we refine iff:

$$(|\mathbf{p_t}| > \mathbf{n_{min}}) \wedge (\mathbf{v_{p_t}} > \mathbf{v_{threshold}})$$

Further, each refinement for a triangle replaces such triangle with at least 2 subtriangles whose pixel-sets form a nontrivial partition of $\mathbf{p_t}$. Since it was shown in **Lemma 1** that the pixel sets form a partition of $\mathbf{P} = \mathbf{\Omega}$, and each refinement strictly increases the number nonempty triangles by definition, the refinement can only perform a finite amount of steps upper-bounded by $|\mathbf{\Omega}|$ in the worst case.

□

Lemma 3. ***Triangulation Invariant***
During and after refinement, triangles still forms a partition of $\mathbf{P} = \mathbf{\Omega}$ *and its exterior edges form the convex hull of P.*

Proof. Let T be the current collection of triangles forming a triangulation of P, and let $\mathbf{t} \in \mathbf{T}$ be a triangle with vertices $\mathbf{v1}, \mathbf{v2}, \mathbf{v3}$ and centroid $\mathbf{c}$. For the sake of example, we split t into 3 subtriangles by the centroid method described in the algorithm.

Then $\mathbf{t'_1}, \mathbf{t'_2}, \mathbf{t'_3}$ have vertices $(\mathbf{v1}, \mathbf{v2}, \mathbf{c}), (\mathbf{v2}, \mathbf{v3}, \mathbf{c}), (\mathbf{v3}, \mathbf{v1}, \mathbf{c})$ respectively, and we obtain the updated $\mathbf{T'} = (\mathbf{T}\backslash\{\mathbf{t}\}) \cup \{\mathbf{t'_1}, \mathbf{t'_2}, \mathbf{t'_3}\}$. By construction, each $\mathbf{t'_i}$ lies entirely inside t, i.e.: $\mathbf{t} = \mathbf{t'_1} \cup \mathbf{t'_2} \cup \mathbf{t'_3}$. Since T was a triangulation of P, $\mathbf{P} = \bigcup_{\mathbf{x} \in \mathbf{T}} \mathbf{x} = (\bigcup_{\mathbf{x} \in \mathbf{T}\backslash\{\mathbf{t}\}} \mathbf{x}) \cup \mathbf{t}$, and when we replace t with its subtriangles $\mathbf{t'_i}$, it does not change this union.

Finally, the interiors of triangles in $\mathbf{T}\backslash\{\mathbf{t}\}$ were disjoint and remain disjoint from the interiors of any refinement. Therefore, the triangles in T' are still disjoint and their union is still P. Thus refinement triangles still form a partition of P and its exterior edges still form the convex hull of P. □

Lemma 4. ***Homogeneity of Final Triangles***
When refinement terminates (proved via Lemma 2), every triangle passes the variance/homogeneity criteria

Proof. By definition of our algorithm, the refinement terminates when there is no triangle left that satisfies $(|\mathbf{p_t}| > \mathbf{n_{min}}) \wedge (\mathbf{v_{p_t}} > \mathbf{v_{threshold}})$. i.e.:

$$\forall \mathbf{t} \in \mathbf{T'}, \neg(|\mathbf{p_t}| > \mathbf{n_{min}} \wedge \mathbf{v_{p_t}} > \mathbf{v_{threshold}})$$

and by De Morgan's Law this is equivalent to:

$$\forall \mathbf{t} \in \mathbf{T'}, (|\mathbf{p_t}| \leq \mathbf{n_{min}} \vee \mathbf{v_{p_t}} \leq \mathbf{v_{threshold}})$$

Thus for every final triangle t in T', the above condition is true. Taking triangles with $|\mathbf{p_t}| \leq \mathbf{n_{min}}$ as trivially homogeneous, by my definition and by statistical intuition of 1-2 pixels not meaningfully describing variance structure, the final triangulation is homogeneous. □

Algorithm Pt.3: Constrained Agglomerative Merging

Input: The refined triangulation: **T'**, the labels: $\mathbf{l_t}$, and a similarity threshold $\delta > \mathbf{0} \in [0, 1]$

Output: A partition of **T'** into regions, **R**, with a label for each region $\mathbf{l_R} \in [0, 1]$

Preliminaries:

Let **R** be an initial collection of regions (triangles) from the refined triangulation. i.e:

$$\mathbf{R} = \{\mathbf{r_t} \ ; \mathbf{t} \in \mathbf{T'}\}$$

where $\mathbf{r_t}$ contains a single triangle from **T'** and has a label $\mathbf{l_R}$ ($l_t = l_R$ here).

Region Adjacency Graph (RAG) and Dissimilarity:

a. Construct a region adjacency graph G = (V,E) where vertices (V) represent regions, and edges (E) represent adjacency.

b. For all neighboring regions i,j, define a dissimilarity matrix $\mathbf{d(i, j)} = |\mathbf{label(i)} - \mathbf{label(j)}|$. If (i,j) *are not* adjacent, set d(i,j) = $+\infty$

Agglomerative Merging:

Let $\mathbf{R} = \{\mathbf{R_1}, \mathbf{R_2}, ..., \mathbf{R_M}\}$ be the current set of regions from T' with label $\mathbf{l_R}$, and let $\sim$ denote that a random pairing R_i, R_j are adjacent. i.e $R_i \sim R_j$

a. Compute the dissimilarity matrix D as:
For $i \neq j$:

$$D_{ij} = \begin{cases} |l_{R_i} - l_{R_j}|, & \text{if } R_i \sim R_j, \\ +\infty, & \text{otherwise.} \end{cases}$$

and $D_{ii} = +\infty$ for all i

b. While $\mathbf{min_{(i,j)}} \mathbf{D_{ij}} \leq \delta$:

(a) Choose a pair of neighboring regions (i',j'), such that: (i',j') = $\mathbf{argmin_{i,j}} D_{ij}$

(b) Merge $R_{i'}$, $R_{j'}$ into K where $\mathbf{K} = \mathbf{R_{i'}} \cup \mathbf{R_{j'}}$, and $l_k = \frac{n_{R_{i'}} l_{R_{i'}} + n_{R_{j'}} l_{R_{j'}}}{n_{R_{i'}} + n_{R_{j'}}}$

(c) Update D by removing i' and j' and add K

(d) For every remaining region R_k in R,

$$D_{Kk} = \begin{cases} |l_K - l_{R_k}|, & \text{if } K \sim R_k, \\ +\infty, & \text{otherwise.} \end{cases}$$

c. *Iterate until the while condition terminates.* **Return** the updated set R and its corresponding labels l_R

Lemma 5. ***Termination of Agglomerative Merging***
The agglomerative merging procedure performs only a finite set of merges and must terminate.

Proof. Let R be the initial collection of regions prior to any merging. As such, every region $\mathbf{r} \in \mathbf{R}$ represents exactly one refined triangle from the termination of refinement. Let n" be the number of initial regions $|\mathbf{R}|$. By construction, each merge takes two distinct adjacent regions, r and s with dissimilarity $\mathbf{d}(\mathbf{r}, \mathbf{s}) \leq \delta$ and replaces them by their union $\mathbf{k} = \mathbf{r} \cup \mathbf{s}$.

Therefore, before the merge $|R| = n''$ for some $\mathbf{n''} \geq \mathbf{2} \in \mathbb{Z}$, and after the merge these regions are removed and one new region is added. Thus, after each merge, the new collection R' has $|\mathbf{R}'| = \mathbf{n''} - \mathbf{1}$.

In other words, every merge strictly decreases the number of regions by 1. Starting from n", the algorithm can perform at most $\mathbf{n''} - \mathbf{1}$ merges since the lowest possible number is 1 (the entire domain). This is the worst case. At best, the triangles after refinement contain no adjacent regions that have dissimilarity $\leq \delta$, or the algorithm performs some merging and terminates once no adjacent pairs have dissimilarity $\leq \delta$. In either cases, because each merge is reducing the number of regions, the agglomerative merging procedure must terminate after finitely many steps. □

Lemma 6. ***Region Partition***
Once the agglomerative merging procedure terminates, the resulting regions still form a partition of P and form its convex hull.

Proof. At the beginning of the agglomerative merging step, each region is defined by a single triangle from the refinement stage's termination. The triangles, as proved in Lemma 3, form a partition of P and their union is the convex hull of P.

Each merge takes two disjoint, adjacent regions i and j, and replaces them by their union $k = i \cup j$. Thus, after any sequence of merges, every region is essentially still a union of triangles from the original triangulation, and every region remains disjoint because unions of disjoint sets are also disjoint. Therefore, at any stage of merging, the union of all regions is P and forms the convex hull of P. No triangles are ever removed, but merging essentially consists of removing/pruning edges from triangles that I have already shown to form a partition of P. □

Discussion:

Applications:

My algorithm provides a unique approach to the segmentation of medical x-rays. I trace my algorithm from a prior over-segmentation, a refinement step, and finally to the merging of similar regions. This approach could potentially allow researchers and clinicians to identify anomalous regions in medial diagnostics — potentially improving the downstream diagnostic capabilities of classifiers.

Considerations/Limitations

As a result of the theoretical nature of this paper, I have a few considerations and limitations to delineate.

1. ***Seeding Points for the Initial Triangulation:*** In my algorithm, I take $\mathbf{P} = \mathbf{\Omega}$ which considered one seed point per-pixel and constructs a Delaunay Triangulation over that grid. However, it may be preferable in certain contexts to be aware of

the placement of the points seeding. For example in medical x-rays, we could do edge-aware seeding or points constrained to a particular region of interest (ROI). Different choices for this seeding affect the geometry of the triangulation and can affect tbe final segmentation as well. As such, it is prudent that I discuss this as a consideration for future work.

2. ***The Choice of*** $\mathbf{v_{threshold}}$: The refinement step (pt.2 of my algorithm) is motivated heavily by a variance threshold ($\mathbf{v_{threshold}}$) which controls what the algorithm considers homogeneous for the triangles. If $\mathbf{v_{threshold}}$ is too high, heterogeneous triangles could be left unrefined and the final triangulation could bring together two completely distinct regions. Since our input image $I \in [0,1], v_{threshold}$ must also live within that range. Since there is no "right" choice for this, cross-validation and hyperparameter tuning are essential here.
3. ***The Choice of*** $\mathbf{n_{min}}$: The choice of n_{min} could also affect the final segmentation. In the algorithm, we treat regions/ triangles with $|\mathbf{p_t}| \leq \mathbf{n_{min}}$ as trivially homogeneous. A smaller value allows triangles with very few pixels to be refined (if their variance is greater than the threshold), which could lead to finer regions (and segmentation) but comes at the cost of complexity and unstable variance estimates. A larger number could result in regions that blur out detail entirely. Thus, the choice of $\mathbf{n_{min}}$ balances resolution against robustness.
4. ***Greedy Nature:*** The final segmentation operates on local, adjacency-constrained agglomerative clustering methods. The greedy nature of this step could lead to dependencies with ties.
5. ***Algorithmic Implementation:*** The biggest consideration is the overall implementation of the algorithm. The overall theoretical nature of the algorithm leads practical uses to be desired/ discovered. This would definitely be the next steps of the study. I would also like to incorporate other types of medical images that include different channels.

Conclusion:

Overall, my algorithm stands to provide insightful information about regions in medical x-rays. Further research must be done to fully validate the algorithm and address the limitations and considerations discussed above.

References:

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